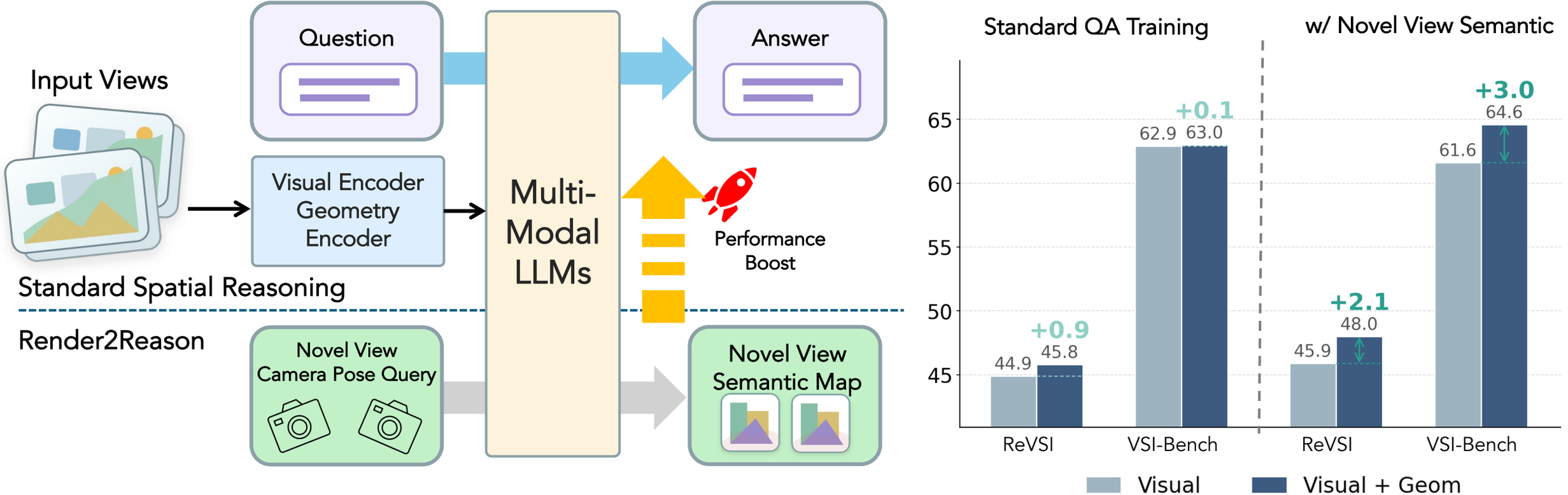

Input Views
Question
Answer
Visual Encoder
Geometry
Encoder
Multi-
Modal
LLMs
Performance
Boost
Standard Spatial Reasoning
Render2Reason
Novel View
Camera Pose Query
Novel View
Semantic Map
Standard QA Training
w/ Novel View Semantic
65
60
55
50
45
+0.9
44.9
45.8
+0.1
62.9
63.0
+2.1
45.9
48.0
+3.0
61.6
64.6
ReVSI
VSI-Bench
ReVSI
VSI-Bench
Visual
Visual + Geom